

Solution Requirements

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>[Users log in via email/password with JWT session management and bcrypt password hashing.]</i>	High
2	Authorization levels	<i>[Three roles defined: Admin (full access), Analyst (predictions & reports), and Viewer (read-only).]</i>	High

3	External interfaces	<i>[Integrates with IBM Cloud Storage, OpenWeatherMap API, and Flask REST API for predictions.]</i>	High
4	Transactions processing	<i>[Weather inputs are sent to the XGBoost model via Flask API, returning a flood risk result within 2 seconds.]</i>	High
5	Reporting	<i>[Generates prediction logs, regional risk dashboards, and exportable CSV/PDF reports.]</i>	Medium
6	Business rules, etc.	<i>[Flood risk classified as High/Medium/Low; predictions below 70% confidence are flagged.]</i>	High
7	Compliance to laws or regulations	<i>[Complies with India's IT Act 2000, DPDP Act 2023, NDMA guidelines, and WCAG 2.1 AA.]</i>	Medium
8	Other	<i>[Admin can swap ML models (DT, RF, KNN, XGBoost) without redeployment; audit logs kept for 12 months.]</i>	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>[System must return prediction results with minimal delay during normal and peak usage.]</i>	<i>Page load < 2 sec; API response < 500 ms</i>
2	Scalability	<i>[Must support multiple concurrent users across regions during peak monsoon season.]</i>	<i>Up to 500 concurrent users; IBM Cloud auto-scaling</i>
3	Security & Data Privacy	<i>[Data must be encrypted in transit and at rest; common vulnerabilities must be addressed.]</i>	<i>HTTPS/TLS 1.2+; AES-256 at rest; OWASP Top 10 covered</i>
4	Reliability & Availability	<i>[System must stay operational during disaster events with automated failure recovery.]</i>	<i>99.5% uptime/month; MTTR < 30 min</i>
5	Usability & Accessibility	<i>[Interface must be accessible to non-technical users on standard devices and browsers.]</i>	<i>WCAG 2.1 AA; Chrome, Firefox, Edge; 1024px+ responsive</i>

6	Other	<i>[ML pipeline must support modular model updates; code must be portable and documented.]</i>	<i>Docker-compatible; PEP 8; Swagger docs; retrain < 4 hrs</i>
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